Three papers, one projector map

A VISUAL READING GUIDE

The author's story: finite distance ⟶ local geometry and holonomy ⟶ signed area and topology

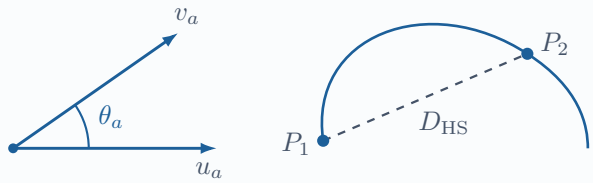
Conceptual order, not chronology

$P(\mathbf{k}) = \Psi(\mathbf{k})\Psi(\mathbf{k})^\dagger : M \longrightarrow \mathrm{Gr}(r, N)$ rank-r occupied subspace $g_{\mu\nu} = \frac{1}{2}\,\mathrm{tr}(\partial_\mu P\,\partial_\nu P)$

H1 DISTANCES

Compare whole subspaces

Occupied states span a subspace; rotating their basis changes no physics. Principal angles compare subspaces. Projector distances reconstruct their sampled image in Euclidean space. Distance data alone do not fix the map's orientation.



one principal pair projector image

$$s_a(\Psi_1^\dagger \Psi_2) = \cos\theta_a$$

$$D_{\mathrm{HS}}^2 = \tfrac{1}{2}\,\mathrm{tr}(P_1 - P_2)^2 = \textstyle\sum_a \sin^2\theta_a$$

H1 §II, Eqs. (4), (8); §III.1.1.

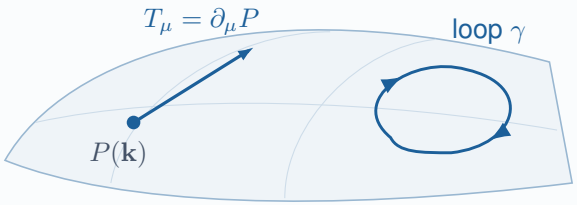
Global view

How far apart are the subspaces?

H2 PROJECTOR CALCULUS

Differentiate and transport

The projector carries local geometry. Its derivatives are tangents; the quantum geometric tensor supplies metric and Berry curvature. Loops produce holonomy. Principal angles determine geodesic length; an ambient geodesic can leave the image.



A closed loop can rotate the frame.

$$q_{\mu\nu} = \mathrm{tr}(P\,\partial_\mu P\,\partial_\nu P) = g_{\mu\nu} - \tfrac{i}{2}\Omega_{\mu\nu}$$

$$\Omega_{\mu\nu} = i\,\mathrm{tr}(P[\partial_\mu P, \partial_\nu P])$$

H2 §IV.2, Eqs. (27)–(30); §IV.5. Geodesics: §IV.3–IV.4, Eq. (61).

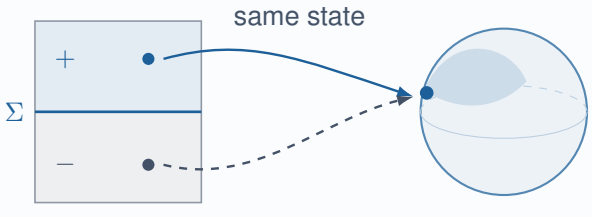
Local and loop view

How does the subspace move?

H3 SIGNED AREA

Account for folds and signs

Two-band states lie on a sphere. Folds make the metric degenerate and overlap sheets with opposite orientations. Unsigned area adds sheets; signed area cancels them. Singular curvature enters Gauss–Bonnet along the fold.



two sheets sphere S^2

$$dA = \tfrac{1}{2}|\Omega|\,d^2k, \qquad d\bar{A} = \tfrac{1}{2}\Omega\,d^2k$$

$$C = \tfrac{1}{2\pi}\textstyle\int_M \Omega\,d^2k = \tfrac{1}{4\pi}\int_M K\,d\bar{A}$$

$K = 4$ on regular regions (H3 §II.2).
H3 §§III–IV, Eqs. (24), (30), (37).

OPEN: MULTIBAND IDENTIFICATION
Extending this geometric link: H3 §V.

Conventions. This poster uses $g = \frac{1}{2}\,\mathrm{tr}(dP\,dP)$ and $\Omega = \Omega_{xy}$ with the H2 sign. H2 and H3 use $g_H = 2g$; H3's displayed Berry curvature is $-\Omega$ here. Signed area is oriented consistently with Ω.

Scope. $P(M)$ is the physical image. H2 §III.2 concerns $U(N)$ group curvature (Eqs. 13–14). H3's singular terms depend on its fold/cusp setting.

H1 Huang and Giataganas, *Exploring Grassmann manifolds in topological systems via quantum distance*, arXiv:2412.20046v2.

H2 Huang, *Quantum-State Projectors on Grassmannian: Geometry, Holonomy, and Topology*, arXiv:2608.06777v1.

H3 Huang, *A Gauss-Bonnet Theorem for Quantum States: Gauss Curvature and Topology in the Projective Hilbert Space*, arXiv:2510.15760v1.